

Plant single-cell solutions for energy and the environment

Benjamin Cole¹, Dominique Bergmann², Crysten E. Blaby-Haas^{3,4},
Ian K. Blaby^{1,5}, Kristofer E. Bouchard^{6,7,8}, Siobhan M. Brady^{9,10},
Doina Ciobanu¹, Devin Coleman-Derr¹¹, Samuel Leibo¹²,
Jenny C. Mortimer^{5,13,14}, Tatsuya Nobori¹⁵, Seung Y. Rhee¹⁶,
Jeremy Schmutz^{1,17}, Blake A. Simmons^{6,13}, Anup K. Singh¹⁸,
Neelima Sinha⁹, John P. Vogel^{1,5,19}, Ronan C. O'Malley¹,
Axel Visel^{1,5} & Diane E. Dickel⁵

Progress in sequencing, microfluidics, and analysis strategies has revolutionized the granularity at which multicellular organisms can be studied. In particular, single-cell transcriptomics has led to fundamental new insights into animal biology, such as the discovery of new cell types and cell type-specific disease processes. However, the application of single-cell approaches to plants, fungi, algae, or bacteria (environmental organisms) has been far more limited, largely due to the challenges posed by polysaccharide walls surrounding these species' cells. In this perspective, we discuss opportunities afforded by single-cell technologies for energy and environmental science and grand challenges that must be tackled to apply these approaches to plants, fungi and algae. We highlight the need to develop better and more comprehensive single-cell technologies, analysis and visualization tools, and tissue preparation methods. We advocate for the creation of a centralized, open-access database to house plant single-cell data. Finally, we consider how such efforts should balance the need for deep characterization of select model species while still capturing the diversity in the plant kingdom. Investments into the development of methods, their application to relevant species, and the creation of resources to support data dissemination will enable groundbreaking insights to propel energy and environmental science forward.

¹ DOE Joint Genome Institute, Lawrence Berkeley National Laboratory, Berkeley, CA, USA. ² Department of Biology, Stanford University and Howard Hughes Medical Institute, Stanford, CA, USA. ³ Biology Department, Brookhaven National Laboratory, Upton, NY, USA. ⁴ Department of Biochemistry and Cell Biology, Stony Brook University, Stony Brook, NY, USA. ⁵ Environmental Genomics and Systems Biology Division, Lawrence Berkeley National Laboratory, Berkeley, CA, USA. ⁶ Biological Systems and Engineering Division, Lawrence Berkeley National Laboratory, Berkeley, CA, USA. ⁷ Computational Research Division, Lawrence Berkeley National Laboratory, Berkeley, CA, USA. ⁸ Helen Wills Neuroscience Institute & Redwood Center for Theoretical Neuroscience, University of California, Berkeley, CA, USA. ⁹ Department of Plant Biology, University of California, Davis, CA, USA. ¹⁰ Genome Center, University of California, Davis, CA, USA. ¹¹ Plant Gene Expression Center, USDA ARS, Albany, CA, USA. ¹² Department of Botany and Plant Pathology, Oregon State University, Corvallis, OR, USA. ¹³ Joint BioEnergy Institute, Emeryville, CA, USA. ¹⁴ School of Agriculture, Food and Wine, Waite Research Institute, Waite Research Precinct, University of Adelaide, Glen Osmond, SA, Australia. ¹⁵ Plant Biology Laboratory and Howard Hughes Medical Institute, Salk Institute for Biological Studies, La Jolla, CA, USA. ¹⁶ Carnegie Institution for Science, Department of Plant Biology, Stanford, CA, USA. ¹⁷ HudsonAlpha Institute for Biotechnology, Huntsville, AL, USA. ¹⁸ Sandia National Laboratories, Livermore, CA, USA. ¹⁹ Department of Plant and Microbial Biology, University of California, Berkeley, CA, USA. ✉email: romalley@lbl.gov; avisel@lbl.gov; dedickel@lbl.gov